

Axler Exercise 3C.12: Associativity of Matrix Multiplication

A Proof and Conceptual Commentary

Linear Algebra Done Right, 4th Edition

Context in the Text

This is Exercise 12 in Section 3C (*Matrices*) of *Linear Algebra Done Right*, 4th edition. Section 3C has just introduced matrix multiplication and motivated its definition via composition of linear maps (Theorem 3.43). The exercise asks the reader to prove associativity of matrix multiplication, and explicitly invites a proof in the spirit of Emil Artin’s remark:

“It is my experience that proofs involving matrices can be shortened by 50% if one throws the matrices out.”

The point of the exercise is to illustrate that matrix multiplication was defined *precisely so that* it mirrors composition of linear maps, and that properties of matrices are best proved by leveraging that correspondence rather than by index-chasing.

Prerequisite

- **Theorem 3.43 (Matrix of a product):** If $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, then

$$\mathcal{M}(ST) = \mathcal{M}(S) \mathcal{M}(T).$$

Statement

Theorem (Exercise 3C.1 — Associativity of matrix multiplication). *Suppose A , B , C are matrices whose sizes are such that $(AB)C$ makes sense. Then $A(BC)$ also makes sense and*

$$(AB)C = A(BC).$$

Why $A(BC)$ Makes Sense

Suppose C is an $m \times p$ matrix, B is an $n \times m$ matrix, and A is a $q \times n$ matrix, so that AB is $q \times m$ and $(AB)C$ is $q \times p$. Then BC is $n \times p$ and $A(BC)$ is $q \times p$ as well. Both products are defined and have the same size.

Equivalently, in operator language: if $C = \mathcal{M}(R)$, $B = \mathcal{M}(T)$, $A = \mathcal{M}(S)$ for linear maps $R \in \mathcal{L}(U, V)$, $T \in \mathcal{L}(V, W)$, $S \in \mathcal{L}(W, Z)$, then both $(AB)C$ and $A(BC)$ represent the matrix of the composite $S \circ T \circ R \in \mathcal{L}(U, Z)$, which is well defined.

Proof

Every matrix is the matrix of some linear map with respect to standard bases. So write $A = \mathcal{M}(S)$, $B = \mathcal{M}(T)$, $C = \mathcal{M}(R)$ for appropriate linear maps R , T , S .

Operator composition is associative by definition — $(ST)R$ and $S(TR)$ both mean “first apply R , then T , then S ” — so:

$$(ST)R = S(TR).$$

Applying $\mathcal{M}$ to both sides and using Theorem 3.43 twice on each side:

$$(AB)C = \mathcal{M}(ST) \mathcal{M}(R) = \mathcal{M}((ST)R) = \mathcal{M}(S(TR)) = \mathcal{M}(S) \mathcal{M}(TR) = A(BC).$$

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Commentary

Note on the Proof Strategy. The proof has two layers, and it is worth separating them clearly.

1. **Operator composition is trivially associative.** There is nothing to prove here: $(ST)R$ and $S(TR)$ are the same function by definition. Associativity is a property inherited for free from the associativity of function composition.
2. **The matrix correspondence $\mathcal{M}$ respects composition.** This is Theorem 3.43, which is non-trivial — it is the entire motivation for why matrix multiplication is defined the way it is. Once you have it, any algebraic identity about composition of operators descends automatically to the same identity about their matrices.

This is Artin’s point in action. A direct proof by indices would require writing out the (i, k) -entry of $(AB)C$ as $\sum_j (\sum_\ell A_{i\ell} B_{\ell j}) C_{jk}$, rearranging the sum using commutativity of scalar multiplication, and matching it against the (i, k) -entry of $A(BC)$. That argument is correct but unmotivating: it does not explain *why* associativity holds, only that the algebra works out. The operator proof, by contrast, makes the reason immediately visible: matrices inherit associativity from functions, via $\mathcal{M}$.